Europe Tourism Flows vs GDP — Business Requirements Document

Oct 7, 2026 · Crafted by @Ramesh

A Tableau dashboard testing whether richer Europeans travel more, and whether they mostly travel to nearby countries, using only free public data.

1. Project objectives

Prepared by: Ramesh · **Role:** Business Analyst (owner, analyst and developer)

The project tests two hypotheses and surfaces the travel patterns behind them.

#	Hypothesis	Question the dashboard answers
H1 — Wealth	Countries with higher GDP per capita send more tourists abroad per resident	Do richer countries travel more?
H2 — Distance	Tourists mostly visit nearby countries, and arrivals fall as distance grows	Does distance limit where people go?

Pattern recognition goals

- **Neighbour pull:** find country pairs with flows far above what wealth and distance predict (e.g., shared border or language).
- **Rich-to-cheaper flows:** check whether richer countries favour lower-GDP neighbours (price effect) or similar-GDP neighbours.
- **Corridors:** identify the top origin–destination pairs and how their share changed from 2010 to 2024.
- **Outliers and shocks:** flag countries that break the pattern, and show the COVID drop (2020–2022) separately.

Business value: a portfolio piece that demonstrates the full BA cycle — hypothesis framing, data sourcing, requirements, analysis and a published dashboard.

2. Scope

In scope	Out of scope
EU-27 plus Norway, Iceland and Switzerland as destinations (where Eurostat reports them)	Countries outside Europe as destinations
Origins: the same European countries	Long-haul origins (US, China, India) — phase 2 idea
Years 2010–2024, with 2020–2022 flagged as COVID years	Forecasting or causal modelling
Yearly arrivals at tourist accommodation, by origin	Day trips, stays with friends or family not in accommodation stats
GDP per capita (PPP) and distance between capitals	Government tourism spending (H3, parked: no free clean dataset)
One Tableau workbook, three dashboard pages	Live or automatic data refresh

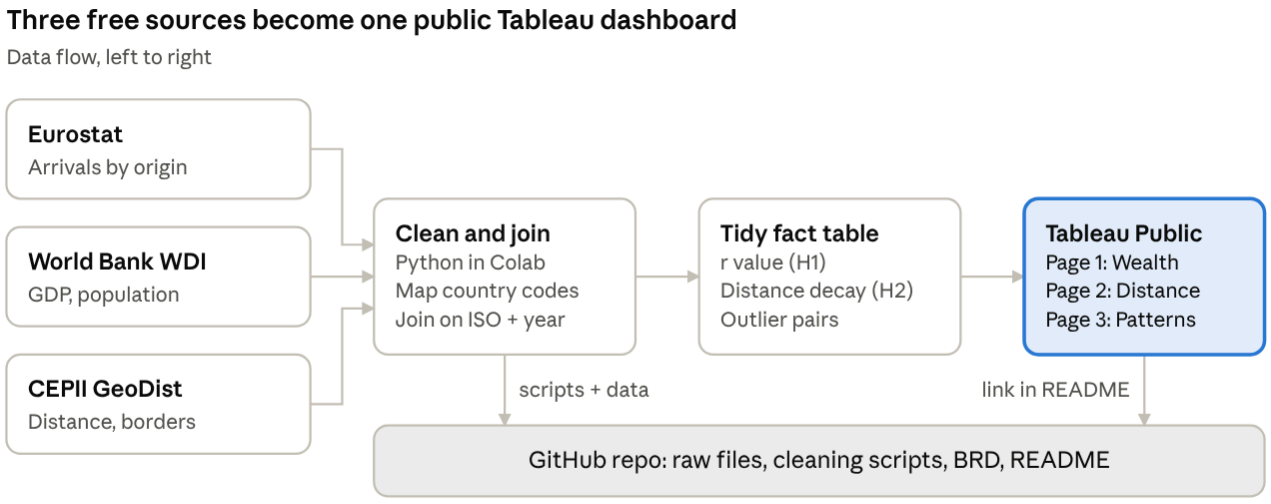
3. Data sources

All three are free; the join key is the ISO country code plus year.

Source	Dataset	Fields used	Grain
Eurostat	tour_occ_arnraw — arrivals at tourist accommodation by country of residence	Destination, origin, year, arrivals	Destination × origin × year
World Bank WDI	GDP per capita, PPP; population	Country, year, GDP per capita, population	Country × year
CEPII GeoDist	Bilateral distances	Origin, destination, distance (km), shared border, common language	Country pair

Caveat shown on every page: Eurostat counts check-ins at accommodation, so one traveller visiting two hotels counts twice, and stays with friends are missed.

4. Schematic flow



data flow · 3 sources, 3 steps, 1 repo

Raw files are cleaned and joined once in Python, analysed into one tidy table, and published as a three-page Tableau Public dashboard; GitHub keeps every file and script.

5. KPIs and measurables

Dashboard KPIs

KPI	Formula	Answers
Outbound trips per 1,000 residents	Arrivals abroad from origin ÷ origin population × 1,000	H1
GDP–travel correlation (r)	Pearson correlation of GDP per capita vs outbound trips per 1,000	H1
Share within 1,000 km	Arrivals to destinations < 1,000 km ÷ all arrivals from that origin	H2
Distance decay	Slope of log(arrivals) vs log(distance)	H2
Corridor share	Arrivals on one origin→destination pair ÷ all arrivals in Europe	Patterns
GDP gap of destination	Destination GDP per capita ÷ origin GDP per capita	Rich-to-cheaper pattern
Change vs 2019	Arrivals this year ÷ arrivals 2019 − 1	Recovery after COVID

Project measurables (definition of success)

- H1 and H2 each get a clear verdict on the dashboard: supported, partly supported or not supported, with the r value or slope shown.
- At least 3 named patterns (corridors or outliers) are called out with a one-line explanation.
- Data covers at least 25 European countries for 2010–2024.
- Published on Tableau Public and linked from resume and LinkedIn.

6. Requirements

Functional (three dashboard pages)

ID	Page	Requirement	Priority
FR-01	1. Wealth (H1)	Scatter of GDP per capita vs outbound trips per 1,000, with trend line and r value	Must
FR-02	1. Wealth	Year slider; play button to animate 2010–2024	Should
FR-03	2. Distance (H2)	Flow map of origin→destination arrivals, line width = arrivals	Must
FR-04	2. Distance	Scatter of distance vs arrivals (log scale), border pairs highlighted	Must
FR-05	2. Distance	Select an origin country to see its top 10 destinations and share within 1,000 km	Must
FR-06	3. Patterns	Top 15 corridors with change vs 2019	Must
FR-07	3. Patterns	Outlier table: pairs far above or below the trend	Should
FR-08	All	KPI tiles at top; filters for year and country; tooltips with all KPIs	Must
FR-09	All	Footer with sources, as-of date and caveats; navigation buttons	Must

Non-functional

- Each page loads in under 5 seconds on Tableau Public.
- Colour-blind-safe palette; every chart labelled.
- Designed for 1366 × 768 desktop; phone layout for page 1.
- Clean data files under 10 MB; every calculated field named and commented.

7. Hosting (all free)

What	Platform	Why
Dashboard	Tableau Public	Free, shareable link, embeddable; workbooks are public by design
Data, cleaning scripts, BRD	GitHub public repo	Version history; shows the work behind the dashboard
Data cleaning	Python (pandas) in Google Colab	Free; repeatable steps
Portfolio write-up	GitHub Pages or LinkedIn article	Links dashboard, repo and BRD in one place

8. Project management

Solo project, 4 working days.

Day	Work	Deliverable
1	BRD sign-off; download Eurostat, World Bank and CEPII data	Signed BRD; raw files in GitHub
2	Clean, map country codes, join; run analysis	Tidy fact table, data dictionary; r value, distance-decay slope, outlier list
3	Build the three dashboard pages	Tableau workbook
4	Test against source, publish, write up	Tableau Public link, GitHub README

Risks

Risk	Mitigation
Missing origin data for some country–year pairs	Show gaps as blank, not zero; note coverage per country
Country codes differ across sources (e.g., EL vs GR for Greece)	Build one country-code mapping table
Correlation read as causation	State clearly on each page: “patterns, not proof”
COVID years distort trends	Toggle to exclude 2020–2022

Acceptance criteria

- [ ] All Must requirements visible in the published workbook
- [ ] Arrivals for 3 sample country pairs match Eurostat to the unit
- [ ] H1 and H2 verdicts shown with their numbers
- [ ] Sources, as-of date and caveats on every page
- [ ] Public links work in a logged-out browser